

Now let $u = [x^2, y^2 + y]$ and $v = y^2 + y$. Then

$$\begin{aligned} [u, v] &= \text{ad}(y^2 + y)^2(x^2) \\ &= (\text{ad}(y)^2 + \text{ad}(y))^2(x^2) \\ &= (\text{ad}(y)^4 + \text{ad}(y)^2)(x^2) \\ &= \text{ad}(y)(x^2) \\ &= [x^2, y], \end{aligned}$$

and $u + [u, v] = [x^2, y^2]$. Plugging all this into (6), we have

$$[[x^2, y^2], \text{ad}([x^2, y])^3([x^2, y^2 + y])] = 0. \quad (7)$$

Now

$$\begin{aligned} \text{ad}([x^2, y])^3([x^2, y^2 + y]) &= \text{ad}([x^2, y])^2([x^2, y], [x^2, y^2] + [x^2, y]) \\ &= \text{ad}([x^2, y])^3([x^2, y^2]) \\ &= \text{ad}([x^2, y])^3 \text{ad}(y)^2(x^2) \\ &= \text{ad}([x^2, y])^3 \text{ad}(y)([x^2, y]) \\ &= \text{ad}([x^2, y])^3[[x^2, y], y]) \\ &= \text{ad}([x^2, y])^4(y) \\ &= \text{ad}([x^2, y])^2(y) + \text{ad}([x^2, y])(y) \\ &= \text{ad}([x^2, y]) \text{ad}(y)^2(x^2) + \text{ad}(y)^2(x^2) \\ &= [[x^2, y], [x^2, y^2]] + [x^2, y^2]. \end{aligned}$$

Thus the left side of (7) simplifies to

$$[[x^2, y^2], [[x^2, y], [x^2, y^2]] + [x^2, y^2]] = [[x^2, y^2], [[x^2, y], [x^2, y^2]]] = \text{ad}([x^2, y^2])^2([x^2, y]).$$

Therefore (7) reduces to

$$\text{ad}([x^2, y^2])^2([x^2, y]) = 0$$

for all $x, y \in R$. By Lemma 3.3,

$$0 = \text{ad}([x^2, y^2])([x^2, y]) = \text{ad}([x^2, y]) \text{ad}(y)^2(x^2) = \text{ad}([x^2, y])^2(y).$$

By Lemma 3.3 again, $\text{ad}([x^2, y])(y) = 0$, that is, $\text{ad}(y)^2(x^2) = 0$. By Lemma 3.3 again, $\text{ad}(y)(x^2) = [x^2, y] = \text{ad}(x)^2(y) = 0$. Applying Lemma 3.3 one last time, $[x, y] = 0$. Therefore R is commutative. $\square$

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